

MATH 310, Section 7.1 Solutions

2. $A = \begin{bmatrix} 1 & -4 \\ -2 & 8 \end{bmatrix}$

$$|\lambda I - A| = \begin{vmatrix} \lambda - 1 & 34 \\ 2 & \lambda - 8 \end{vmatrix} = (\lambda - 1)(\lambda - 8) - 2 \cdot 4 = \lambda^2 - 9\lambda = \lambda^2 - 9\lambda.$$

(a) Therefore the characteristic polynomial is $\lambda^2 - 9\lambda$ and the characteristic equation is $\lambda^2 - 9\lambda = 0$

(b) Because $0 = \lambda^2 - 9\lambda = \lambda(\lambda - 9)$, we see that the eigenvalues are 0 and 9.

(c) **Eigenvector for 0** is $\mathbf{x}$ for which $A\mathbf{x} = 0\mathbf{x}$, that is, $\begin{bmatrix} x \\ y \end{bmatrix}$ with $\begin{bmatrix} 1 & -4 \\ -2 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

Solving in the usual way, $\begin{bmatrix} 1 & -4 & 0 \\ -2 & 8 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -4 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, so $x = 4y$.

Thus the eigenvectors are $\begin{bmatrix} 4y \\ y \end{bmatrix} = y \begin{bmatrix} 4 \\ 1 \end{bmatrix}$, with $y \in \mathbb{R}$. So the eigenspace for 0 is the span of $\begin{bmatrix} 4 \\ 1 \end{bmatrix}$.

(d) **Eigenvector for 9** is $\mathbf{x}$ for which $A\mathbf{x} = 9\mathbf{x}$, that is, $\mathbf{x}$ for which $A\mathbf{x} - 9\mathbf{x} = \mathbf{0}$, or $(A - 9I)\mathbf{x} = \mathbf{0}$.

$$\text{Now, } (A - 9I) = \begin{bmatrix} 1 & -4 \\ -2 & 8 \end{bmatrix} - \begin{bmatrix} 9 & 0 \\ 0 & 9 \end{bmatrix} = \begin{bmatrix} -8 & -4 \\ -2 & -1 \end{bmatrix}.$$

Therefore we are looking for $\begin{bmatrix} x \\ y \end{bmatrix}$ with $\begin{bmatrix} -8 & -4 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

Solving in the usual way, $\begin{bmatrix} -8 & -4 & 0 \\ -2 & -1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$, so $x = -\frac{1}{2}y$.

So the eigenvectors are $\begin{bmatrix} -\frac{1}{2}y \\ y \end{bmatrix} = \frac{1}{2}y \begin{bmatrix} -1 \\ 2 \end{bmatrix}$, with $y \in \mathbb{R}$. So the eigenspace for 9 is the span of $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$.

12. $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & 4 \\ 0 & 0 & 1 \end{bmatrix}$

(a) The characteristic polynomial is $|\lambda I - A| = \begin{vmatrix} \lambda - 2 & 0 & -1 \\ 0 & \lambda - 3 & -4 \\ 0 & 0 & \lambda - 1 \end{vmatrix} = \boxed{(\lambda - 2)(\lambda - 3)(\lambda - 1)}$

(b) The eigenvalues are the roots of the characteristic polynomial, and are thus $\lambda_1 = 1$, $\lambda_2 = 2$, and $\lambda_3 = 3$.

(c) i. Eigenvectors for $\lambda = 1$:

Need to solve $(1I - A)\mathbf{x} = \mathbf{0}$

$$\begin{bmatrix} -1 & 0 & -1 & 0 \\ 0 & -2 & -4 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \text{ The solutions are } x = -z, y = -2z, z = t$$

so the eigenvectors for $\lambda = 1$ are $\begin{bmatrix} -t \\ -2t \\ t \end{bmatrix} = t \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}$

ii. Eigenvectors for $\lambda = 2$:

Need to solve $(2I - A)\mathbf{x} = \mathbf{0}$

$$\begin{bmatrix} 0 & 0 & -1 & 0 \\ 0 & -1 & -4 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \text{ The solutions are } x = t, y = 0, z = 0$$

so the eigenvectors corresponding to $\lambda = 2$ are $\begin{bmatrix} t \\ 0 \\ 0 \end{bmatrix} = t \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$.

iii. Eigenvectors for $\lambda = 3$:

Need to solve $(3I - A)\mathbf{x} = \mathbf{0}$

$$\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & -4 & 0 \\ 0 & 0 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \text{ The solutions are } x = 0, \ y = t, \ z = 0$$

so the eigenvectors corresponding to $\lambda = 3$ are $\begin{bmatrix} 0 \\ t \\ 0 \end{bmatrix} = t \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$
